

Magnetic Signature-Based Model Using Machine Learning for Electrical and Mechanical Faults Classification of Wind Turbine Drive Trains

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Abstract—Signal processing and fault indicators analysis are essential for efficient fault detection, classification, and diagnosis of wind turbines. Accordingly, existing works proposed the installation of multiple intrusive sensors (e.g., current, voltage, and accelerometer) for data collection in order to detect and classify the faults in wind turbine drive trains (WTDT). However, these sensors are scattered on the drive train and have a limited local reach on its components making it technically difficult to install. Therefore, signals from these sensors are not able to detect multi parameter phenomena such as coupling of the mechanical and electrical components of the drive train which contains essential fault information. This work proposes the use of magnetic signatures as fault condition indicators of the complete drive train due to the ability of contactless measurement of this signal without opening the main components of the drive train. This is achieved by performing non-destructive magnetic modeling and analysis of the entire drive train. The air gap magnetic flux density of the wind generator is demonstrated as a good fault condition indicator for different common faults occurring on the gearbox, bearings, and the generator. The proposed model is validated using a supervised machine learning classification algorithm in a way to distinguish between electrical and mechanical faults.

Index Terms—Condition Monitoring Systems, Fault indicators, Magnetic modeling, Magnetic Signature, Wind turbine drivetrains.

I. INTRODUCTION

Rapid deployment of wind energy is essential to meet the federal and state goals for sustainable energy development. However, investments in wind energy hugely depend on the competitive electricity prices of various energy sources including maintenance cost, fault detection, and restoration of damaged components [1]. Therefore, there is an urgent need to reduce the cost of wind technologies to make them viable for long term investments. Currently, the monitoring and maintenance paradigm in wind turbines is shifting from

a reactive, periodic and usage based maintenance to a predictive, regular and condition based maintenance [2]. This requires continuous monitoring and fault prognosis of various components. Typical data sources considered for condition monitoring in wind turbine drive trains are based on supervisory control and data acquisition (SCADA) data (e.g., current, voltage, frequency), vibration monitoring, acoustics monitoring, temperature monitoring, and current signature analysis. However, there are certain disadvantages to these existing monitoring techniques. Most of these techniques except acoustic monitoring requires intrusive sensors [3]. The invasiveness of these sensors makes them difficult for installation. Further, these installations can cause structural abrasions and interfere with the flow of current in certain components, increasing their remaining useful life. Thus, they add an extra cost during replacement and upgrading of components and the monitoring is restricted only to certain individual components. Therefore, mechanical and electrical faults have the same effects on these measurements making the classification of faults a complicated task and requiring the use of additional sensing such as the magnetic field variable. Therefore, data on dynamic and interdependent variables of the wind turbines are generally not available from these sensors data.

Wind turbine drive trains are most vulnerable to faults due to the presence of dynamic components (e.g., electrical, mechanical) that are coupled to each other and hence its downtime poses a huge impact on the cost-effectiveness of the installation [4]. Further, the presence of electromagnetic coupling among its components generates dynamic parameters (e.g., magnetic signature, electromagnetic torque) that can be considered fault signatures. However, they are neglected during individual component sensing. Therefore, current literatures have acknowledged the need for efficient modeling of drive trains to capture these dynamic parameters [5]. This paper proposes to employ magnetic modeling of drive trains to obtain dynamic